

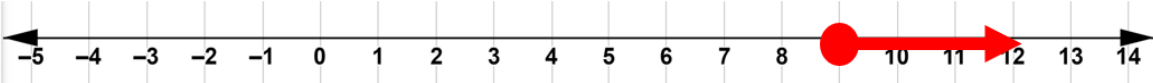
6th Grade Accelerated Math Unit 14 Assessment Guide

General Information	The benchmarks on this assessment are closely related yet distinctive in their expectations. For example, MA.6.AR.1.2 and MA.7.AR.2.1 both relate to representing mathematical and real-world contexts using inequalities but the 6th grade benchmark does not require students to perform an operation to solve while they 7th grade benchmark does. Both MA.6.AR.2.2 and MA.6.AR.2.3 have students write equations. The cognitive demand is similar for both of these benchmarks. Interpreting real-world contexts can be difficult for students. Providing them with a variety of contexts and encouraging students to draw models and/or pictures to help support their thinking process can help students move to the abstract. When solving equations or inequalities, students may show work using algebraic steps, models, or pictures. Clarification 1 of benchmark MA.6.AR.2.4 states, "Instruction focuses on using algebraic reasoning, drawings, and mental math to determine unknowns." Questions 1 and 8 are set up so that the student is using algebraic reasoning or mental math to solve. The benchmarks on this assessment have assessment limits that may be helpful for the teacher to understand. For example, in the test item specifications, MA.6.AR.1.2 has an assessment limit that the given context should be continuous. This means that the contexts should not be true for just certain values as seen in the example in the benchmark, "Mrs. Anna told her class that they will get a pizza if the class has an average of at least 83 out of 100 correct questions on the semester exam. The inequality $g \geq 83$ can be used to represent the situation where students receive a pizza and the inequality $g < 83$ can be used to represent the situation where students do not receive a pizza." This context is not continuous because, as described, each question is correct or incorrect so only whole numbers are possible. This example would not be a type of problem students would see on the statewide assessment. Fluency of operations with rational numbers is an important outcome of Grade 6 Accelerated Math. Teachers should consider how to monitor simple calculation errors. If teachers want to consider student fluency of rational numbers, then if a student makes a simple calculation error that is carried through their work, the students receive partial credit. Students may inadvertently write the wrong variable for an answer. This is a common mistake and should receive a minimal point deduction, if any. It is suggested that if a student does it more than twice, 1 point is deducted for each occurrence past two.
Calculator Information	The benchmarks assessed on the unit assessment are either calculator neutral, which means that a calculator does not give a student an advantage or disadvantage when showing their knowledge of the intended benchmark, or no calculator. Based on they fluency expectations for this course, it is recommended students do not have a calculator for this assessment.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.6.AR.2.4
	The value of h can be determined to be ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ because ☐ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☒ $\frac{7}{7}$ ☐ $\frac{52}{7}$ is ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ ☐ less than ☒ more than ☐ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ OR The value of h can be determined to be ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ because ☒ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ is ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ ☒ less than ☐ more than ☐ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☒ $\frac{37}{7}$ ☐ $\frac{52}{7}$	Recommended Scoring (10 Total Points)	Consider giving 2 points for each correct response and deducting 2 points for each incorrect response box.

Question	2	Benchmark(s)	MA.6.AR.2.3
$x = -18$		Recommended Scoring (4 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $x = 18$ if the student simply lost the negative sign.
Question	3	Benchmark(s)	MA.6.AR.2.2
$b = 135$		Recommended Scoring (4 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points if the student simply lost the negative sign.
Question	4	Benchmark(s)	MA.6.AR.2.2
$w = -11$		Recommended Scoring (4 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $w = 11$ if the student simply lost the negative sign.

Question	5	Benchmark(s)	MA.6.AR.2.3
$h = -225$		Recommended Scoring (4 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $h = 225$ if the student simply lost the negative.
Question	6a	Benchmark(s)	MA.6.AR.2.2
Equation $x - 112 = 204$		Recommended Scoring (4 Total Points)	Students may choose to write the equation as $x - 204 = 112$ or $x = 204 + 112$. Students should not lose points for correct mathematics.
Question	6b	Benchmark(s)	MA.6.AR.2.2
316 minutes		Recommended Scoring (4 Total Points)	Students may get this correct and miss question 11a.
Question	7	Benchmark(s)	MA.6.AR.2.3
$\frac{c}{75} = 4$		Recommended Scoring (4 Total Points)	Consider giving no partial credit.

Question	8	Benchmark(s)	MA.6.AR.2.4
The value of h in the bar model is -2.1. 5. -10.5. The bar model represents the equation $-2.1h = -5.$ $\frac{h}{5} = -2.1.$ $h = -2.1.$ Recommended Scoring (8 Total Points) Consider giving 4 points for each correctly completed statement.			
Question	9	Benchmark(s)	MA.6.AR.2.2
Equation $x + 36 = 110$		Recommended Scoring (4 Total Points)	Students may write the answer as $x = 110 - 36$ or $110 - x = 36$.
Question	10	Benchmark(s)	MA.6.AR.1.2
$d \geq 9$ 		Recommended Scoring (6 Total Points)	Consider giving students 3 points for the correct inequality and 3 points for the correct graph.

Question	11	Benchmark(s)	MA.7.AR.2.1
$x > -21.6$		Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the open circle, and 2 points for the correct ray. Students may write -21.6 using equivalent values.
		Additional Notes	This is the only question where reversing the inequality sign is assessed. This is because if students fail to do that once, it is likely they will repeat the same error. Grade 6 Accelerated students should be able to distinguish when to reverse the sign. Students who reverse the inequality sign on questions 11 and 13 have a misunderstanding about when to reverse the sign.

Question	12	Benchmark(s)	MA.7.AR.2.1
$w > -\frac{47}{6}$ 	Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the open circle, and 2 points for the correct ray. Students may write $-\frac{47}{6}$ using equivalent values.	
	Additional Notes	Students may make calculation errors, which may point to a fluency issue with rational numbers.	
Question	13	Benchmark(s)	MA.7.AR.2.1
$-144 \leq y$ 	Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the closed circle, and 2 points for the correct ray.	
	Additional Notes	Students who reverse the inequality sign on question 13 have a misunderstanding about when to reverse the sign. Students may correctly solve the inequality but draw the ray in the incorrect direction.	

Question	14	Benchmark(s)	MA.6.AR.1.2
$\frac{1200}{12}$ m 76.3 m 76.2 m $74\frac{5}{9}$ m 76.05 m 76.5 m $76\frac{1}{8}$ m $\frac{7620}{1000}$ m $76\frac{12}{16}$ m $\frac{1500}{20}$ m		Recommended Scoring (12 Total Points)	Consider giving students 2 points for the correct value. Consider deduction of 1 point for each incorrect value.
Question	15	Benchmark(s)	MA.7.AR.2.1
15a Inequality $17.50x \leq 112$ 15b $x \leq 6.4$		Recommended Scoring (8 Total Points)	Suggested points is 5 points for correctly writing the inequality and 3 points for correctly solving the inequality that was written. Students should not lose 8 points if they correctly solve an incorrect inequality that they wrote for part A.